Scientific Calculator using Java Swing

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1. Introduction

The Scientific Calculator is a desktop-based application developed using Java Swing. It provides both basic arithmetic operations and advanced scientific functions through a graphical user interface (GUI). The project demonstrates the implementation of object-oriented programming concepts and event-driven programming in Java.

2. Objectives

- To develop a GUI-based scientific calculator using Java Swing.
- To understand event-driven programming in Java.
- To implement mathematical operations using Java Math library.
- To apply Object-Oriented Programming (OOP) concepts.

3. Technologies Used

- Java (JDK 25)
- Swing (GUI Framework)
- AWT (Layout and UI Support)
- Event Handling (ActionListener)
- Java Math Library

4. Features

- Basic arithmetic operations (+, -, *, /)
- Trigonometric functions (sin, cos, tan)
- Square and Square Root functions
- Logarithm (Base 10)
- Clear and Exit buttons
- Modern Dark User Interface

5. System Design

The system consists of the following main components: JFrame (main window), JTextField (display screen), JPanel (button container), JButton (input keys), and ActionListener (event handling mechanism). The layout managers used are BorderLayout and GridLayout for organizing the components effectively.

6. Working Mechanism

When a user clicks any button, the ActionListener detects the event and executes the actionPerformed() method. Based on the button pressed, the program performs the corresponding mathematical operation using the Java Math library and displays the result in the text field.

7. Conclusion

The Scientific Calculator project successfully demonstrates the implementation of GUI programming, event handling, and object-oriented principles in Java. It enhances understanding of desktop application development and mathematical function integration using the Java programming language.